Math 105: Calculus with Algebra
Instructor: David Zureick-Brown ("DZB")

All assignments

Gradescope code: B5J8EN

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Gradescope Instructions for submitting work in Math 105

You will be using the online Gradescope progam to submit your homework and exams. These instructions tell you how to sign up initially, and how to submit your written work.

Signing up for Gradescope the first time.

If you haven't used Gradescope for an **Amherst College** course before:

- Go to `http://www.gradescope.com`, click on "Sign up for free" (which may auto-scroll you to the bottom of the page), and select Sign up as [a] "Student".
- In the signup box:
  - Use the course entry code **B5J8EN**
  - Use your full name
  - Use your **Amherst College email** address. Or, if you are a Five-College student, use your email address from your own school.
  - Leave the "Student ID" entry blank.
- You will probably get an email asking to set a password for your account, so check your `amherst.edu` email inbox. (Or your email inbox through your own school, for Five-College students.)

Adding Math 105 to Gradescope.

If you **have** used Gradescope for an Amherst course before, and so you already have an account through your `amherst.edu` email, you still need to add Math 105, so:

- Go to `http://www.gradescope.com` and log in.
- Go to your Account Dashboard (click the Gradescope logo at upper left), and click "Add Course" at bottom right.
- Use the course code **B5J8EN**

(submission instructions on next page)

Submitting written work

First write it out on paper as you would normally. Then **scan it** to create a PDF. One method for scanning is the smartphone app **DropBox**. It makes nice clear scans, and it saves them directly into a folder so that you can have all your assignments in one place. **CamScanner** is another free scanning App, and there are others, too. **Gradescope** now has its own scanning app. You can also use a printer/scanner if you prefer.

Please be kind to our dear graders and make sure your submission is legible !

In particular, please leave some spacing between separate problems.

If you have a tablet computer, you may write your work there (instead of on paper) and save it as a PDF.

Some of you may know the math formatting package LaTeX and may want to use it in Math 105. That's fine, too; if so, you may write up your work in LaTeX and save the resulting PDF.

In short, any method is fine as long as it creates a legible **PDF** file and **NOT a photo**.

For example, if you use the DropBox app, then in your created *Math 105 Homework* Dropbox folder, you can select create (+) at the bottom of the screen and click the *Scan Document* option. Snap a shot of the first page of your homework, and then click [+] to snap shots of any subsequent pages. Do **not** use the *Take Photo* option.

After you have scanned/saved your work as a PDF, submit it on Gradescope as follows:

- Go to `http://www.gradescope.com` and log in.
- Select the course "Math 105, Fall 2026" and the appropriate assignment.
- Select "submit pdf" to submit your work in PDF format. Browse to find your PDF and upload.
- Now it is time to **tag** your problems. This is an **important step** , where you are telling Gradescope which problems are on which page(s).

  For each problem, select the pages of your submission where your written solution appears.

  I think the easiest thing to do is to click on the page of **your** homework upload where you wrote the given problem, and then click on the assigned problem listed. Repeat for each problem.

  You must tag the problems or else you will not get credit for your work.

  Gradescope will give you a warning when you go to submit your assignment if you have not selected the pages correctly. If you tag a problem incorrectly, you can fix it by clicking "More" and "Reselect Pages".
- Click Save or Submit.

After your assignment is graded, you will be able to see your score on the written problems, along with comments, on Gradescope. You should receive an email notifying you when each homework set is graded.

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 1

Due **Friday, September 6** at 11:55pm, on Gradescope

Goals:

- Review of basic algebra.
- Review of absolute values and inequalities.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer. For a review on inequalities and absolute value, see Appendix A (Stewart Calculus book).

(1) Simplify:

$$4 - (5 - 3) + 3(4 - 7) - 4(1 + 2)$$

(2) In the following expressions, simplify and reduce to lowest terms. (Some answers may have more than one form)

(a)

$$\frac{\frac{4}{75}}{\frac{8}{25}}$$

(b)

$$\frac{7x}{3y} \cdot \frac{3y + 2}{x}$$

(c)

$$\frac{\frac{xy}{(x+y)}}{\frac{x^2y}{(x+y)^3}}$$

MORE PROBLEMS ON THE NEXT PAGE →

(3) Express as a single fraction and simplify:

(a)
$$\frac{1}{2}+\frac{4}{3}-\frac{2}{5}-\frac{3}{15}$$

(b)
$$\frac{\frac{1}{3}+\frac{2}{5}}{\frac{3}{2}}$$

(c)
$$\frac{1}{x}-\frac{x+1}{xy}+\frac{x-2}{xz}$$

(d)
$$\frac{\frac{1}{st}-\frac{1}{w}}{\frac{1}{tw}-\frac{2}{s}}$$

(4) Rewrite the expression without using the absolute-value symbol:

$$|x-2| \text{ if } x<2$$

(5) Solve the inequalities with the **geometric interpretation** of absolute value.

(a)
$$|x|<3$$

(b)
$$|x-4|<1$$

(c)
$$|x+5|\geq 2$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 1 SOLUTIONS

(1) Simplify:

$$4-(5-3)+3(4-7)-4(1+2)$$

Solution. We simplify the expressions inside parentheses first:

$$4-(5-3)+3(4-7)-4(1+2)=4-2+3(-3)-4(3).$$

Therefore

$$4-2-9-12=-19.$$

Hence the simplified value is $\boxed{-19}$.

(2) In the following expressions, simplify and reduce to lowest terms. (Some answers may have more than one form)

(a)

$$\frac{\frac{4}{75}}{\frac{8}{25}}$$

Solution. Dividing by a fraction is the same as multiplying by its reciprocal:

$$\frac{\frac{4}{75}}{\frac{8}{25}}=\frac{4}{75}\cdot\frac{25}{8}=\frac{1}{3}\cdot\frac{1}{2}=\boxed{\frac{1}{6}}.$$

(b)

$$\frac{7x}{3y}\cdot\frac{3y+2}{x}$$

Solution. Cancel the common factor x:

$$\frac{7x}{3y}\cdot\frac{3y+2}{x}=\boxed{\frac{7(3y+2)}{3y}}.$$

(c)

$$\frac{\frac{xy}{(x+y)}}{\frac{x^2y}{(x+y)^3}}$$

Solution. Multiply by the reciprocal of the denominator:

$$\frac{\frac{xy}{x+y}}{\frac{x^2y}{(x+y)^3}}=\frac{xy}{x+y}\cdot\frac{(x+y)^3}{x^2y}.$$

Cancelling common factors gives

$$\boxed{\frac{(x+y)^2}{x}}.$$

(3) Express as a single fraction and simplify:

(a)

$$\frac{1}{2}+\frac{4}{3}-\frac{2}{5}-\frac{3}{15}$$

Solution. A common denominator is 30:

$$\frac{1}{2}+\frac{4}{3}-\frac{2}{5}-\frac{3}{15}=\frac{15}{30}+\frac{40}{30}-\frac{12}{30}-\frac{6}{30}=\boxed{\frac{37}{30}}.$$

(b)

$$\frac{\frac{1}{3}+\frac{2}{5}}{\frac{3}{2}}$$

Solution. First simplify the numerator:

$$\frac{1}{3}+\frac{2}{5}=\frac{5}{15}+\frac{6}{15}=\frac{11}{15}.$$

Then divide by 3/2:

$$\frac{\frac{11}{15}}{\frac{3}{2}}=\frac{11}{15}\cdot\frac{2}{3}=\boxed{\frac{22}{45}}.$$

(c)

$$\frac{1}{x}-\frac{x+1}{xy}+\frac{x-2}{xz}$$

Solution. The common denominator is xyz. Thus

$$\frac{1}{x}-\frac{x+1}{xy}+\frac{x-2}{xz}=\frac{yz}{xyz}-\frac{z(x+1)}{xyz}+\frac{y(x-2)}{xyz}.$$

Combining the numerators,

$$\boxed{\frac{xy+yz-xz-2y-z}{xyz}}.$$

(d)

$$\frac{\frac{1}{st}-\frac{1}{w}}{\frac{1}{tw}-\frac{2}{s}}$$

Solution. Put the numerator and denominator over the common denominator stw:

$$\frac{1}{st}-\frac{1}{w}=\frac{w-st}{stw},\qquad \frac{1}{tw}-\frac{2}{s}=\frac{s-2tw}{stw}.$$

Therefore

$$\frac{\frac{w-st}{stw}}{\frac{s-2tw}{stw}}=\boxed{\frac{w-st}{s-2tw}}.$$

(4) Rewrite the expression without using the absolute-value symbol:

$$|x-2| \text{ if } x<2$$

Solution. If $x<2$, then $x-2<0$. The absolute value of a negative number is its opposite, so

$$|x-2|=-(x-2)=\boxed{2-x}.$$

(5) Solve the inequalities with the **geometric interpretation** of absolute value.

(a)

$$|x| < 3$$

Solution. The quantity $|x|$ is the distance from x to 0. We want that distance to be less than 3, so x must lie between -3 and 3:

$$\boxed{-3 < x < 3}.$$

(b)

$$|x - 4| < 1$$

Solution. The quantity $|x - 4|$ is the distance from x to 4. We want that distance to be less than 1, so

$$\boxed{3 < x < 5}.$$

(c)

$$|x + 5| \geq 2$$

Solution. Since $|x + 5| = |x - (-5)|$, this is the distance from x to -5. We want that distance to be at least 2. Thus x must be at least 2 units to the left or right of -5:

$$\boxed{x \leq -7 \quad \text{or} \quad x \geq -3}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 2

Due **Monday, September 7** at 11:55pm, on Gradescope

Goals:

- Review equations of lines.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer. For a review on equations of lines, see Appendix A (Stewart Calculus book).

Find the slope of the line through P and Q:

(1) $P(1, 5), Q(4, 11)$

(2) $P(-1, -4), Q(6, 0)$

Find the equation of the line that satisfies the given conditions.

(3) Through $(2, -3)$ with slope 6

(4) Through $(2, 1)$ and $(1, 6)$

(5) Slope 3, y-intercept -2

(6) x-intercept 1, y-intercept -3

(7) Through $(4, 5)$, parallel to x-axis

(8) Through $(4, 5)$, parallel to y-axis

(9) Through $(1, -6)$, parallel to the line $x + 2y = 6$

(10) Through $(-1, -2)$, perpendicular to the line $2x + 5y + 8 = 0$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 2 SOLUTIONS

Find the slope of the line through P and Q:

(1) $P(1,5), Q(4,11)$
Solution. The slope through two points (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Thus

$$m = \frac{11-5}{4-1} = \frac{6}{3} = \boxed{2}.$$

(2) $P(-1,-4), Q(6,0)$
Solution. Using the slope formula,

$$m = \frac{0-(-4)}{6-(-1)} = \frac{4}{7}.$$

Therefore the slope is

$$\boxed{\frac{4}{7}}.$$

Find the equation of the line that satisfies the given conditions.

(3) Through $(2,-3)$ with slope 6
Solution. Using point-slope form $y - y_1 = m(x - x_1)$,

$$y - (-3) = 6(x-2).$$

Hence

$$y + 3 = 6x - 12,$$

so

$$\boxed{y = 6x - 15}.$$

(4) Through $(2,1)$ and $(1,6)$
Solution. First find the slope:

$$m = \frac{6-1}{1-2} = \frac{5}{-1} = -5.$$

Using the point $(2,1)$,

$$y - 1 = -5(x-2).$$

Therefore

$$y - 1 = -5x + 10,$$

so

$$\boxed{y = -5x + 11}.$$

(5) Slope 3, y-intercept -2

Solution. A line with slope m and y-intercept b has equation $y = mx + b$. Here $m = 3$ and $b = -2$, so

$$\boxed{y = 3x - 2}.$$

(6) x-intercept 1, y-intercept -3

Solution. The x-intercept gives the point $(1, 0)$, and the y-intercept gives the point $(0, -3)$. The slope is

$$m = \frac{0 - (-3)}{1 - 0} = 3.$$

Since the y-intercept is -3, the equation is

$$\boxed{y = 3x - 3}.$$

(7) Through $(4, 5)$, parallel to x-axis

Solution. A line parallel to the x-axis is horizontal, so its y-coordinate is constant. Since the line passes through $(4, 5)$,

$$\boxed{y = 5}.$$

(8) Through $(4, 5)$, parallel to y-axis

Solution. A line parallel to the y-axis is vertical, so its x-coordinate is constant. Since the line passes through $(4, 5)$,

$$\boxed{x = 4}.$$

(9) Through $(1, -6)$, parallel to the line $x + 2y = 6$

Solution. Rewrite the given line in slope-intercept form:

$$x + 2y = 6 \implies 2y = -x + 6 \implies y = -\frac{1}{2}x + 3.$$

Its slope is $-\frac{1}{2}$. A parallel line has the same slope, so using point-slope form with $(1, -6)$,

$$y + 6 = -\frac{1}{2}(x - 1).$$

Therefore

$$\boxed{y = -\frac{1}{2}x - \frac{11}{2}}.$$

Equivalently, the equation can be written as $\boxed{x + 2y = -11}$.

(10) Through $(-1,-2)$, perpendicular to the line $2x+5y+8=0$

Solution. First rewrite the given line in slope-intercept form:

$$2x+5y+8=0 \implies 5y=-2x-8 \implies y=-\frac{2}{5}x-\frac{8}{5}.$$

Its slope is $-\frac{2}{5}$. A perpendicular line has slope equal to the negative reciprocal, so its slope is

$$\frac{5}{2}.$$

Using point-slope form with $(-1,-2)$,

$$y+2=\frac{5}{2}(x+1).$$

Thus

$$y+2=\frac{5}{2}x+\frac{5}{2},$$

and therefore

$$\boxed{y=\frac{5}{2}x+\frac{1}{2}}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 3

Due **Friday, September 11** at 11:55pm, on Gradescope

Goals:

- Review of the essential ideas behind functions.
- Practice graphing of functions.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer.

(1) Let $f(x) = x^3 + 2x$. Determine:

  (a) $f(2)$

  (b) $f(2x)$

  (c) $f(-x)$

  (d) $f(2 + \Delta x)$

(2) Let $u(t) = \frac{1}{2}t^2 + t$. Determine:

  (a) $u(-1)$

  (b) $u(t + \Delta t)$

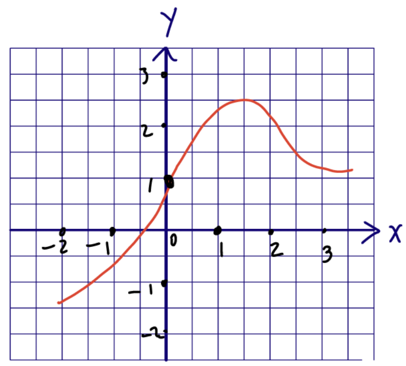
  (c) $u(-t)$

(3) Sketch the graph of the function $g(t) = |1 - 3t|$

(4) Sketch the graph of the function

$$f(x) = \begin{cases} |x| & \text{if } |x| \leq 1 \\ 1 & \text{if } |x| > 1 \end{cases}$$

(5) The graph of the function f is given.



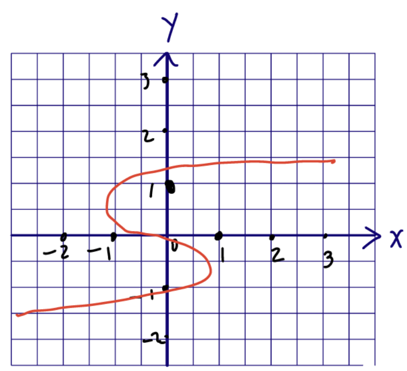
(a) State the value of $f(1.5)$.

(b) Estimate the value of $f(-1)$.

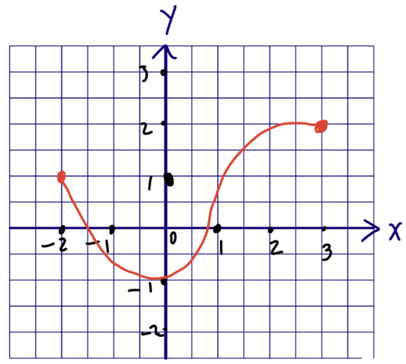
(c) For what values of x is $f(x) = 1$?

(d) Estimate the value of x such that $f(x) = 0$.

(6) Determine whether the curve is a graph of a function of x. If it is, state the domain and the range of the function.



(7) State the domain and the range of the function graphed below.



(8) Find the domain of $f(x) = \frac{x+4}{x^2-9}$

(9) Match the each equation with its graph. Explain your choices. Don't use a computer or graphic calculator. (a) $y = x^2$ (b) $y = x^5$ (c) $y = x^8$